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Exercise Sheet 7

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Give an example of an infinite group in which every nontrivial subgroup is infinite.

Exercise 2.
Show that if $a^2 = e$ for all elements a in a group G, then G must be abelian.

Exercise 3.
Let G be a group. Show that

$$Z(G) = \{x \in G \mid gx = xg \text{ for all } g \in G\}$$

is a subgroup of G. This subgroup is called the center of G..

Exercise 4.
Prove that the set of matrices of the form

$$\begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix}$$

is a group under matrix multiplication. This group, known as the Heisenberg group[1], is important in quantum physics. Matrix multiplication in the Heisenberg group is defined as

$$\begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x' & y' \\ 0 & 1 & z' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+x' & y+y'+xz' \\ 0 & 1 & z+z' \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 5.
Let G be an abelian group and $\phi : G \to H$ a group homomorphism. Prove that $\phi(G)$ is also abelian.

[1] https://en.wikipedia.org/wiki/Heisenberg_group

Exercise 6.
Let $\phi : G \to H$ be a homomorphism and define a relation $\sim$ on G by $a \sim b$ if and only if $\phi(a) = \phi(b)$ for $a, b \in G$. Show that $\sim$ is an equivalence relation and describe the equivalence classes.

Exercise 7.
Prove or disprove the following assertion. Let G, H, and K be groups. If $G \times K \cong H \times K$, then $G \cong H$.

Exercise 8.
Let $G \cong H$. Show that if G is cyclic, then also H is cyclic.

Hint: *Let g be a generator for G. If $\phi : G \to H$ is an isomorphism, show that $\phi(g)$ is a generator for H.*

★ **Exercise 9.**
Let G be a group. We say that a function $\phi : G \to G$ is an **automorphism** if it is an isomorphism from G to itself. Let $\mathrm{Aut}(G)$ be the set of all automorphism of G. Prove that this set is a group.

★ **Exercise 10.**
An **inner automorphism** of G

$$i_g : G \to G$$

is defined by the map

$$i_g(x) = gxg^{-1}$$

for $g \in G$. Prove that i_g is an automorphism of G.